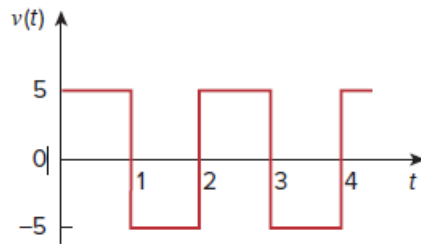


Problem

Determine the rms value of the waveform in Fig. 11.55.

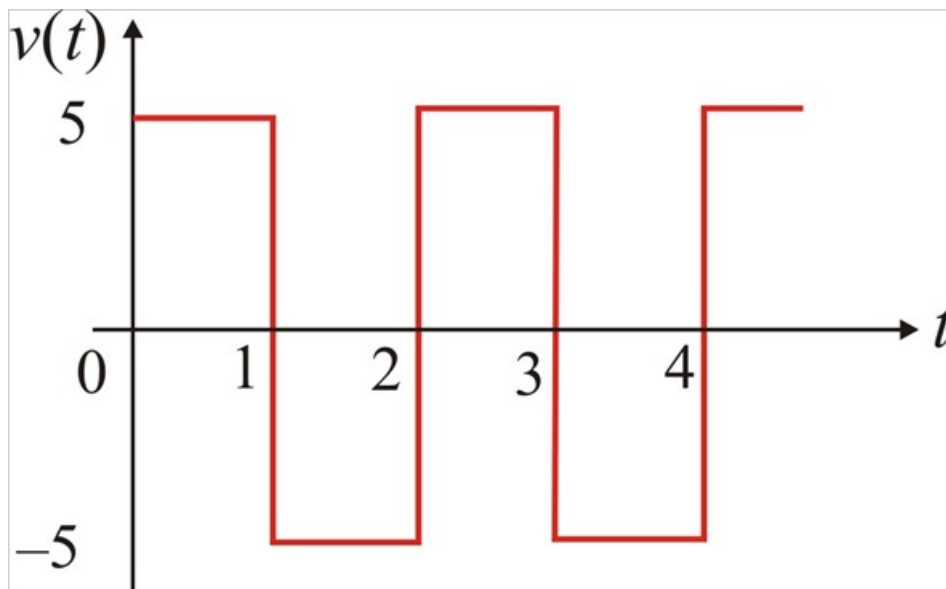
Figure 11.55:



Step-by-step solution

Step 1 of 2

Given voltage waveform



Step 2 of 2

Period of above waveform is $T = 2$

From the above waveform, voltage is defined as

$$v(t) = \begin{cases} 5 & \text{for } 0 < t < 1 \\ -5 & \text{for } 1 < t < 2 \end{cases}$$

$$V_{\text{rms}}^2 = \frac{1}{T} \left[\int_0^T (v(t))^2 dt \right]$$

$$V_{\text{rms}}^2 = \frac{1}{2} \left[\int_0^1 (5)^2 dt + \int_1^2 (-5)^2 dt \right]$$

$$V_{\text{rms}}^2 = \frac{1}{2} [25(1-0) + 25(2-1)]$$

$$V_{\text{rms}}^2 = \frac{1}{2} [50]$$

$$V_{\text{rms}}^2 = 25$$

$$\boxed{V_{\text{rms}} = 5\text{V}}$$